

Мікросервіси з використанням Messaging Queue

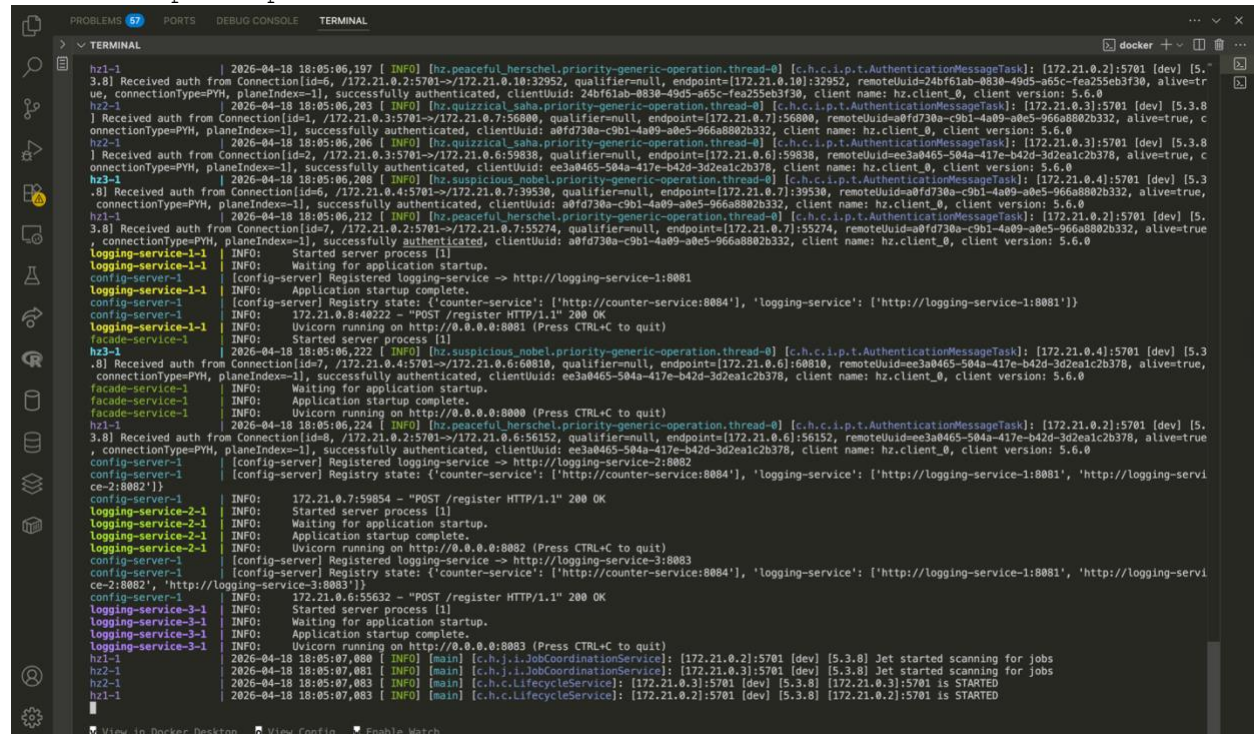
Репозиторій: https://github.com/YOUR_USERNAME/YOUR_REPO/tree/micro_mq

Система складається з наступних компонентів:

- **config-server**. Кожен сервіс при запуску реєструє свою адресу через POST-запит. facade-service звертається до нього перед кожним запитом для отримання актуальних адрес.
- **facade-service**. При POST-запиті логує повідомлення в один з екземплярів logging-service (випадковий вибір) та асинхронно кладе повідомлення у чергу Hazelcast Queue для counter-service. При GET-запиті зчитує логи з logging-service та лічильники з counter-service через HTTP.
- **logging-service** зберігає повідомлення у розподіленій структурі Hazelcast Distributed Map.
- **counter-service** -- фоновий потік постійно читає повідомлення з Hazelcast Queue та оновлює лічильники у базі даних (in-memory dict).
- **Hazelcast cluster** — надає Distributed Map для logging-service та Distributed Queue для асинхронної передачі повідомлень до counter-service.

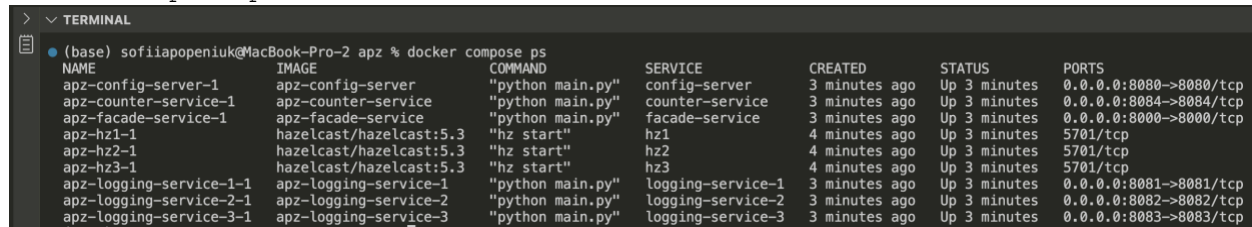
Запуск системи

docker compose up --build



```
h2z-1 | 2026-04-18 18:05:06,197 [ INFO] [hz.peaceful_herschel.priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]: [172.21.0.2]:5701 [dev] [5.3.8] Received auth from ConnectionId=6, /172.21.0.2:5701->/172.21.0.10:32952, qualifier=null, endpoint=[172.21.0.10]:32952, remoteUid=24b6f1ab-0830-49d5-a65c-fea255eb3f30, client name: hz.client_0, client version: 5.6.0, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: 24b6f1ab-0830-49d5-a65c-fea255eb3f30, client name: hz.client_0, client version: 5.6.0
h2z-1 | 2026-04-18 18:05:06,203 [ INFO] [hz.quizzical_saha.priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]: [172.21.0.3]:5701 [dev] [5.3.8] Received auth from ConnectionId=1, /172.21.0.3:5701->/172.21.0.7:56800, qualifier=null, endpoint=[172.21.0.7]:56800, remoteUid=a0fd730a-c9b1-4a09-ae5-966a8802b332, alive=true, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: a0fd730a-c9b1-4a09-ae5-966a8802b332, client name: hz.client_0, client version: 5.6.0
h2z-1 | 2026-04-18 18:05:06,206 [ INFO] [hz.quizzical_saha.priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]: [172.21.0.4]:5701 [dev] [5.3.8] Received auth from ConnectionId=6, /172.21.0.4:5701->/172.21.0.7:39530, qualifier=null, endpoint=[172.21.0.7]:39530, remoteUid=a0fd730a-c9b1-4a09-ae5-966a8802b332, alive=true, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: a0fd730a-c9b1-4a09-ae5-966a8802b332, client name: hz.client_0, client version: 5.6.0
h2z-1 | 2026-04-18 18:05:06,212 [ INFO] [hz.peaceful_herschel.priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]: [172.21.0.2]:5701 [dev] [5.3.8] Received auth from ConnectionId=7, /172.21.0.2:5701->/172.21.0.6:59838, qualifier=null, endpoint=[172.21.0.6]:59838, remoteUid=ee3a0465-504a-417e-b42d-3d2ealc2b378, alive=true, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: ee3a0465-504a-417e-b42d-3d2ealc2b378, client name: hz.client_0, client version: 5.6.0
h2z-1 | 2026-04-18 18:05:06,206 [ INFO] [hz.suspicious_nobel.priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]: [172.21.0.4]:5701 [dev] [5.3.8] Received auth from ConnectionId=2, /172.21.0.2:5701->/172.21.0.7:55274, qualifier=null, endpoint=[172.21.0.7]:55274, remoteUid=a0fd730a-c9b1-4a09-ae5-966a8802b332, alive=true, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: a0fd730a-c9b1-4a09-ae5-966a8802b332, client name: hz.client_0, client version: 5.6.0
logging-service-1-1 | INFO: Started server process [1]
logging-service-1-1 | INFO: Waiting for application startup.
config-server-1 | [config-server] Registered logging-service -> http://logging-service-1:8081
logging-service-1-1 | INFO: Application startup complete.
config-server-1 | [config-server] Registry state: {'counter-service': ['http://counter-service:8084'], 'logging-service': ['http://logging-service-1:8081']}
config-server-1 | INFO: 172.21.0.8:40222 - "POST /register HTTP/1.1" 200 OK
logging-service-1-1 | INFO: Uvicorn running on http://0.0.0.0:8081 (Press CTRL+C to quit)
facade-service-1 | INFO: Started server process [1]
h2z-1 | 2026-04-18 18:05:06,222 [ INFO] [hz.suspicious_nobel.priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]: [172.21.0.4]:5701 [dev] [5.3.8] Received auth from ConnectionId=7, /172.21.0.4:5701->/172.21.0.6:60810, qualifier=null, endpoint=[172.21.0.6]:60810, remoteUid=ee3a0465-504a-417e-b42d-3d2ealc2b378, alive=true, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: ee3a0465-504a-417e-b42d-3d2ealc2b378, client name: hz.client_0, client version: 5.6.0
facade-service-1 | INFO: Waiting for application startup.
facade-service-1 | INFO: Application startup complete.
facade-service-1 | INFO: Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)
h2z-1 | 2026-04-18 18:05:06,224 [ INFO] [hz.peaceful_herschel.priority-generic-operation.thread-0] [c.h.c.i.p.t.AuthenticationMessageTask]: [172.21.0.2]:5701 [dev] [5.3.8] Received auth from ConnectionId=6, /172.21.0.2:5701->/172.21.0.6:56152, qualifier=null, endpoint=[172.21.0.6]:56152, remoteUid=ee3a0465-504a-417e-b42d-3d2ealc2b378, alive=true, connectionType=PYH, planeIndex=-1], successfully authenticated, clientUid: ee3a0465-504a-417e-b42d-3d2ealc2b378, client name: hz.client_0, client version: 5.6.0
config-server-1 | [config-server] Registered logging-service -> http://logging-service-2:8082
config-server-1 | [config-server] Registry state: {'counter-service': ['http://counter-service:8084'], 'logging-service': ['http://logging-service-1:8081', 'http://logging-service-2:8082']}
config-server-1 | INFO: 172.21.0.7:59854 - "POST /register HTTP/1.1" 200 OK
logging-service-2-1 | INFO: Started server process [1]
logging-service-2-1 | INFO: Waiting for application startup.
logging-service-2-1 | INFO: Application startup complete.
config-server-1 | INFO: Uvicorn running on http://0.0.0.0:8082 (Press CTRL+C to quit)
config-server-1 | [config-server] Registered logging-service -> http://logging-service-3:8083
config-server-1 | [config-server] Registry state: {'counter-service': ['http://counter-service:8084'], 'logging-service': ['http://logging-service-1:8081', 'http://logging-service-2:8082', 'http://logging-service-3:8083']}
config-server-1 | INFO: 172.21.0.6:55632 - "POST /register HTTP/1.1" 200 OK
logging-service-3-1 | INFO: Started server process [1]
logging-service-3-1 | INFO: Waiting for application startup.
logging-service-3-1 | INFO: Application startup complete.
h2z-1 | 2026-04-18 18:05:07,080 [ INFO] [main] [c.h.j.l.i.JobCoordinationService]: [172.21.0.2]:5701 [dev] [5.3.8] Jet started scanning for jobs
h2z-1 | 2026-04-18 18:05:07,081 [ INFO] [main] [c.h.j.l.i.JobCoordinationService]: [172.21.0.3]:5701 [dev] [5.3.8] Jet started scanning for jobs
h2z-1 | 2026-04-18 18:05:07,083 [ INFO] [main] [c.h.c.l.LifecycleService]: [172.21.0.3]:5701 [dev] [5.3.8] [172.21.0.3]:5701 is STARTED
h2z-1 | 2026-04-18 18:05:07,083 [ INFO] [main] [c.h.c.l.LifecycleService]: [172.21.0.2]:5701 [dev] [5.3.8] [172.21.0.2]:5701 is STARTED
```

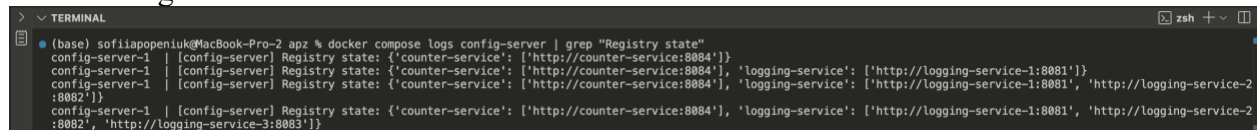
docker compose ps



NAME	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORTS
apz-config-server-1	apz-config-server	"python main.py"	config-server	3 minutes ago	Up 3 minutes	0.0.0.0:8080->8080/tcp
apz-counter-service-1	apz-counter-service	"python main.py"	counter-service	3 minutes ago	Up 3 minutes	0.0.0.0:8084->8084/tcp
apz-facade-service-1	apz-facade-service	"python main.py"	facade-service	3 minutes ago	Up 3 minutes	0.0.0.0:8000->8000/tcp
apz-hz1-1	hazelcast/hazelcast:5.3	"hz start"	hz1	4 minutes ago	Up 3 minutes	5701/tcp
apz-hz2-1	hazelcast/hazelcast:5.3	"hz start"	hz2	4 minutes ago	Up 3 minutes	5701/tcp
apz-hz3-1	hazelcast/hazelcast:5.3	"hz start"	hz3	4 minutes ago	Up 3 minutes	5701/tcp
apz-logging-service-1-1	apz-logging-service-1	"python main.py"	logging-service-1	3 minutes ago	Up 3 minutes	0.0.0.0:8081->8081/tcp
apz-logging-service-2-1	apz-logging-service-2	"python main.py"	logging-service-2	3 minutes ago	Up 3 minutes	0.0.0.0:8082->8082/tcp
apz-logging-service-3-1	apz-logging-service-3	"python main.py"	logging-service-3	3 minutes ago	Up 3 minutes	0.0.0.0:8083->8083/tcp

Реєстрація сервісів у config-server

При старті кожен мікросервіс реєструє себе в config-server через POST-запит. Це видно у логах config-server:



```
(base) sofiaopeniuk@MacBook-Pro-2 apz % docker compose logs config-server | grep "Registry state"
config-server-1 | [config-server] Registry state: {'counter-service': ['http://counter-service:8084']}
config-server-1 | [config-server] Registry state: {'counter-service': ['http://counter-service:8084'], 'logging-service': ['http://logging-service-1:8081']}
config-server-1 | [config-server] Registry state: {'counter-service': ['http://counter-service:8084'], 'logging-service': ['http://logging-service-1:8081', 'http://logging-service-2:8082']}
config-server-1 | [config-server] Registry state: {'counter-service': ['http://counter-service:8084'], 'logging-service': ['http://logging-service-1:8081', 'http://logging-service-2:8082', 'http://logging-service-3:8083']}
```

Запис 10 транзакцій через POST

Через facade-service записуємо 10 повідомлень msg1–msg10:

> ▾ **TERMINAL**



```
--- msg2 ---
{
  "status": "ok",
  "uuid": "faeb5e78-547b-4796-92bf-b5f0849b0f1b",
  "logged_by": "http://logging-service-3:8083"
}
--- msg3 ---
{
  "status": "ok",
  "uuid": "e8b4b215-37b2-4d72-8487-91ba77178f2a",
  "logged_by": "http://logging-service-3:8083"
}
--- msg4 ---
{
  "status": "ok",
  "uuid": "6c437e0b-22f9-4f90-a881-dc66ee490b20",
  "logged_by": "http://logging-service-1:8081"
}
--- msg5 ---
{
  "status": "ok",
  "uuid": "c87e4969-cbdd-442e-b755-1816ddacb2fc",
  "logged_by": "http://logging-service-3:8083"
}
--- msg6 ---
{
  "status": "ok",
  "uuid": "462608ae-ae2d-437c-a8f7-ed2dd6342148",
  "logged_by": "http://logging-service-3:8083"
}
--- msg7 ---
{
  "status": "ok",
  "uuid": "e20d717a-6cc4-4bdc-ab50-ff953c177ff7",
  "logged_by": "http://logging-service-2:8082"
}
--- msg8 ---
{
  "status": "ok",
  "uuid": "f7ffe119-e903-48a2-bbc6-f3c035c840d2",
  "logged_by": "http://logging-service-3:8083"
}
--- msg9 ---
{
  "status": "ok",
  "uuid": "c6280c7d-f660-4d83-a201-54d8e6d9c0f6",
  "logged_by": "http://logging-service-2:8082"
}
--- msg10 ---
{
  "status": "ok",
  "uuid": "deb7bc9d-6450-4479-ad9a-e7e502cb5854",
  "logged_by": "http://logging-service-2:8082"
}
```

Кожна відповідь містить UUID повідомлення та ім'я logging-service екземпляра, який його обробив (logged_by).

Розподіл повідомлень між екземплярами logging-service

Повідомлення розподіляються випадковим чином між трьома екземплярами logging-service.

```
● (base) sofiiapopeniuk@MacBook-Pro-2 apz % echo "=== logging-service-1 ===" && docker compose logs logging-service-1 | grep "Received POST"
echo "=== logging-service-2 ===" && docker compose logs logging-service-2 | grep "Received POST"
echo "=== logging-service-3 ===" && docker compose logs logging-service-3 | grep "Received POST"

=== logging-service-1 ===
logging-service-1-1 | [logging-service:8081] Received POST uuid=98bd476c-0880-4722-b230-16617fe73ca9 msg=msg1
logging-service-1-1 | [logging-service:8081] Received POST uuid=6c437e0b-22f9-4f90-a881-dc66ee490b20 msg=msg4
=== logging-service-2 ===
logging-service-2-1 | [logging-service:8082] Received POST uuid=e20d717a-6cc4-4bdc-ab50-ff953c177ff7 msg=msg7
logging-service-2-1 | [logging-service:8082] Received POST uuid=c6280c7d-f660-4d83-a201-54d8e6d9c0f6 msg=msg9
logging-service-2-1 | [logging-service:8082] Received POST uuid=deb7bc9d-6450-4479-ad9a-e7e502cb5854 msg=msg10
=== logging-service-3 ===
logging-service-3-1 | [logging-service:8083] Received POST uuid=faeb5e78-547b-4796-92bf-b5f0849b0f1b msg=msg2
logging-service-3-1 | [logging-service:8083] Received POST uuid=e8b4b215-37b2-4d72-8487-91ba77178f2a msg=msg3
logging-service-3-1 | [logging-service:8083] Received POST uuid=c87e4969-cbdd-442e-b755-1816ddacb2fc msg=msg5
logging-service-3-1 | [logging-service:8083] Received POST uuid=462608ae-ae2d-437c-a8f7-ed2dd6342148 msg=msg6
logging-service-3-1 | [logging-service:8083] Received POST uuid=f7ffe119-e903-48a2-bbc6-f3c035c840d2 msg=msg8
```

Видно, що різні екземпляри отримали різні повідомлення, що підтверджує роботу балансування навантаження.

Читання транзакцій та перевірка лічильників через GET

```
● (base) sofiiapopeniuk@MacBook-Pro-2 apz % curl -s http://localhost:8000/ | python3 -m json.tool
{
  "logs": {
    "6c437e0b-22f9-4f90-a881-dc66ee490b20": "msg4",
    "deb7bc9d-6450-4479-ad9a-e7e502cb5854": "msg10",
    "c6280c7d-f660-4d83-a201-54d8e6d9c0f6": "msg9",
    "e20d717a-6cc4-4bdc-ab50-ff953c177ff7": "msg7",
    "e8b4b215-37b2-4d72-8487-91ba77178f2a": "msg3",
    "462608ae-ae2d-437c-a8f7-ed2dd6342148": "msg6",
    "f7ffe119-e903-48a2-bbc6-f3c035c840d2": "msg8",
    "98bd476c-0880-4722-b230-16617fe73ca9": "msg1",
    "faeb5e78-547b-4796-92bf-b5f0849b0f1b": "msg2",
    "c87e4969-cbdd-442e-b755-1816ddacb2fc": "msg5"
  },
  "counters": {
    "msg1": 1,
    "msg2": 1,
    "msg3": 1,
    "msg4": 1,
    "msg5": 1,
    "msg6": 1,
    "msg7": 1,
    "msg8": 1,
    "msg9": 1,
    "msg10": 1
  }
}
```

Перевірка відмовостійкості

Зупинка counter-service

```
● (base) sofiiaropeniuk@MacBook-Pro-2 apz % docker compose pause counter-service && echo "counter-service paused"
[+] Pausing 1/1
✓ Container apz-counter-service-1 Paused
counter-service paused
```

Запис транзакцій при недоступному counter-service

POST-запити мають виконуватись успішно — повідомлення накопичуються у черзі:

```
● (base) sofiiaropeniuk@MacBook-Pro-2 apz % for i in $(seq 11 15); do
  echo "--- msg$i ---"
  curl -s -X POST http://localhost:8000/ \
    -H "Content-Type: application/json" \
    -d "{\"msg\": \"msg$i\"}" | python3 -m json.tool
done

--- msg11 ---
{
  "status": "ok",
  "uuid": "02d01ab6-6aff-49da-a053-06f139133491",
  "logged_by": "http://logging-service-1:8081"
}
--- msg12 ---
{
  "status": "ok",
  "uuid": "703c4436-1882-4d01-a5b9-23e2ae486192",
  "logged_by": "http://logging-service-3:8083"
}
--- msg13 ---
{
  "status": "ok",
  "uuid": "47f1c735-32a9-460c-bac9-fbd3cd679659",
  "logged_by": "http://logging-service-1:8081"
}
--- msg14 ---
{
  "status": "ok",
  "uuid": "2533f6f7-8d99-464c-8f39-8686f1b2bd07",
  "logged_by": "http://logging-service-3:8083"
}
--- msg15 ---
{
  "status": "ok",
  "uuid": "6b2ca1ad-302f-47bf-b0a6-ea6b925829b4",
  "logged_by": "http://logging-service-3:8083"
}
```

GET при недоступному counter-service


```
(base) sofiiapopeniuk@MacBook-Pro-2 apz % curl -s http://localhost:8000/ | python3 -m json.tool
{
  "logs": {
    "6c437e0b-22f9-4f90-a881-dc66ee490b20": "msg4",
    "deb7bc9d-6450-4479-ad9a-e7e502cb5854": "msg10",
    "c6280c7d-f660-4d83-a201-54d8e6d9c0f6": "msg9",
    "2533f6f7-8d99-464c-8f39-8686f1b2bd07": "msg14",
    "e20d717a-6cc4-4bdc-ab50-ff953c177ff7": "msg7",
    "e8b4b215-37b2-4d72-8487-91ba77178f2a": "msg3",
    "462608ae-ae2d-437c-a8f7-ed2dd6342148": "msg6",
    "703c4436-1882-4d01-a5b9-23e2ae486192": "msg12",
    "47f1c735-32a9-460c-bac9-fbd3cd679659": "msg13",
    "02d01ab6-6aff-49da-a053-06f139133491": "msg11",
    "f7ffe119-e903-48a2-bbc6-f3c035c840d2": "msg8",
    "98bd476c-0880-4722-b230-16617fe73ca9": "msg1",
    "6b2ca1ad-302f-47bf-b0a6-ea6b925829b4": "msg15",
    "faeb5e78-547b-4796-92bf-b5f0849b0f1b": "msg2",
    "c87e4969-cbdd-442e-b755-1816ddac2fc": "msg5"
  },
  "counters": null
}
```

Відновлення counter-service та обробка накопичених повідомлень

```
docker compose unpause counter-service && echo "counter-service unpaused"
sleep 5
curl -s http://localhost:8000/ | python3 -m json.tool
```

```
> TERMINAL
(base) sofiiapopeniuk@MacBook-Pro-2 apz % docker compose unpause counter-service && echo "counter-service unpaused"
sleep 5
curl -s http://localhost:8000/ | python3 -m json.tool

[+] Running 1/1
  ✓ Container apz-counter-service-1 Unpaused
counter-service unpaused
{
  "logs": {
    "6c437e0b-22f9-4f90-a881-dc66ee490b20": "msg4",
    "deb7bc9d-6450-4479-ad9a-e7e502cb5854": "msg10",
    "c6280c7d-f660-4d83-a201-54d8e6d9c0f6": "msg9",
    "2533f6f7-8d99-464c-8f39-8686f1b2bd07": "msg14",
    "e20d717a-6cc4-4bdc-ab50-ff953c177ff7": "msg7",
    "e8b4b215-37b2-4d72-8487-91ba77178f2a": "msg3",
    "462608ae-ae2d-437c-a8f7-ed2dd6342148": "msg6",
    "703c4436-1882-4d01-a5b9-23e2ae486192": "msg12",
    "47f1c735-32a9-460c-bac9-fbd3cd679659": "msg13",
    "02d01ab6-6aff-49da-a053-06f139133491": "msg11",
    "f7ffe119-e903-48a2-bbc6-f3c035c840d2": "msg8",
    "98bd476c-0880-4722-b230-16617fe73ca9": "msg1",
    "6b2ca1ad-302f-47bf-b0a6-ea6b925829b4": "msg15",
    "faeb5e78-547b-4796-92bf-b5f0849b0f1b": "msg2",
    "c87e4969-cbdd-442e-b755-1816ddac2fc": "msg5"
  },
  "counters": {
    "msg1": 1,
    "msg2": 1,
    "msg3": 1,
    "msg4": 1,
    "msg5": 1,
    "msg6": 1,
    "msg7": 1,
    "msg8": 1,
    "msg9": 1,
    "msg10": 1,
    "msg11": 1,
    "msg12": 1,
    "msg13": 1,
    "msg14": 1,
    "msg15": 1
  }
}
```

Після відновлення counter-service зчитує накопичені в черзі повідомлення (msg11–msg15) та оновлює лічильники. У відповіді з'являються всі 15 лічильників.

Консоль counter-service

```

● (base) sofiapopeniuk@MacBook-Pro-2 apz % docker compose logs counter-service
counter-service-1 | INFO:      Started server process [1]
counter-service-1 | INFO:      Waiting for application startup.
counter-service-1 | INFO:      Application startup complete.
counter-service-1 | INFO:      Uvicorn running on http://0.0.0.0:8084 (Press CTRL+C to quit)
counter-service-1 | [counter-service] Connected to Hazelcast
counter-service-1 | [counter-service] Registered: {'status': 'ok'}
counter-service-1 | [counter-service] Consumer loop started
counter-service-1 | [counter-service] Consumed from queue: msg1
counter-service-1 | [counter-service] Consumed from queue: msg2
counter-service-1 | [counter-service] Consumed from queue: msg3
counter-service-1 | [counter-service] Consumed from queue: msg4
counter-service-1 | [counter-service] Consumed from queue: msg5
counter-service-1 | [counter-service] Consumed from queue: msg6
counter-service-1 | [counter-service] Consumed from queue: msg7
counter-service-1 | [counter-service] Consumed from queue: msg8
counter-service-1 | [counter-service] Consumed from queue: msg9
counter-service-1 | [counter-service] Consumed from queue: msg10
counter-service-1 | [counter-service] GET /counter -> {'msg1': 1, 'msg2': 1, 'msg3': 1, 'msg4': 1, 'msg5': 1, 'msg6': 1, 'msg7': 1, 'msg8': 1, 'msg9': 1, 'msg10': 1}
counter-service-1 | INFO:      172.21.0.10:43134 - "GET /counter HTTP/1.1" 200 OK
counter-service-1 | [counter-service] GET /counter -> {'msg1': 1, 'msg2': 1, 'msg3': 1, 'msg4': 1, 'msg5': 1, 'msg6': 1, 'msg7': 1, 'msg8': 1, 'msg9': 1, 'msg10': 1}
counter-service-1 | [counter-service] Consumed from queue: msg11
counter-service-1 | [counter-service] GET /counter -> {'msg1': 1, 'msg2': 1, 'msg3': 1, 'msg4': 1, 'msg5': 1, 'msg6': 1, 'msg7': 1, 'msg8': 1, 'msg9': 1, 'msg10': 1, 'msg11': 1}
counter-service-1 | [counter-service] Consumed from queue: msg12
counter-service-1 | [counter-service] Consumed from queue: msg13
counter-service-1 | [counter-service] Consumed from queue: msg14
counter-service-1 | [counter-service] Consumed from queue: msg15
counter-service-1 | [counter-service] GET /counter -> {'msg1': 1, 'msg2': 1, 'msg3': 1, 'msg4': 1, 'msg5': 1, 'msg6': 1, 'msg7': 1, 'msg8': 1, 'msg9': 1, 'msg10': 1, 'msg11': 1, 'ms
g12': 1, 'msg13': 1, 'msg14': 1, 'msg15': 1}
counter-service-1 | INFO:      172.21.0.10:43402 - "GET /counter HTTP/1.1" 200 OK

```

Видно:

- підключення до Hazelcast
- реєстрацію в config-server
- старт фонового consumer-поточу
- послідовне зчитування msg1–msg10 з черги
- зчитування msg11–msg15 після відновлення

Консоль facade-service

docker compose logs facade-service

```
> v TERMINAL
● (base) sofiiapopeniuk@MacBook-Pro-2 apz % docker compose logs facade-service
facade-service-1 | INFO: Started server process [1]
facade-service-1 | INFO: Waiting for application startup.
facade-service-1 | INFO: Application startup complete.
facade-service-1 | INFO: Uvicorn running on http://0.0.0.0:8080 (Press CTRL+C to quit)
facade-service-1 | [facade-service] Connected to Hazelcast
facade-service-1 | [facade-service] POST -> logging-service at http://logging-service-1:8081, uuid=98bd476c-0880-4722-b230-16617fe73ca9
facade-service-1 | [facade-service] Enqueuing to counter-queue: msg1
facade-service-1 | INFO: 192.168.65.1:37826 - "POST / HTTP/1.1" 200 OK
facade-service-1 | [facade-service] POST -> logging-service at http://logging-service-3:8083, uuid=faeb5e78-547b-4796-92bf-b5f0849b0f1b
facade-service-1 | [facade-service] Enqueuing to counter-queue: msg2
facade-service-1 | INFO: 192.168.65.1:62505 - "POST / HTTP/1.1" 200 OK
facade-service-1 | [facade-service] POST -> logging-service at http://logging-service-3:8083, uuid=e8b4b215-37b2-4d72-8487-91ba77178f2a
facade-service-1 | [facade-service] Enqueuing to counter-queue: msg3
facade-service-1 | INFO: 192.168.65.1:44308 - "POST / HTTP/1.1" 200 OK
facade-service-1 | [facade-service] POST -> logging-service at http://logging-service-1:8081, uuid=6c437e0b-22f9-4f90-a881-dc66ee490b20
facade-service-1 | [facade-service] Enqueuing to counter-queue: msg4
facade-service-1 | INFO: 192.168.65.1:24515 - "POST / HTTP/1.1" 200 OK
facade-service-1 | [facade-service] POST -> logging-service at http://logging-service-3:8083, uuid=c87e4969-cbdd-442e-b755-1816ddacb2fc
facade-service-1 | [facade-service] Enqueuing to counter-queue: msg5
facade-service-1 | INFO: 192.168.65.1:25342 - "POST / HTTP/1.1" 200 OK
facade-service-1 | [facade-service] POST -> logging-service at http://logging-service-3:8083, uuid=462608ae-ae2d-437c-a8f7-ed2dd6342148
facade-service-1 | [facade-service] Enqueuing to counter-queue: msg6
facade-service-1 | INFO: 192.168.65.1:60016 - "POST / HTTP/1.1" 200 OK
facade-service-1 | [facade-service] POST -> logging-service at http://logging-service-2:8082, uuid=e20d717a-6cc4-4bdc-ab50-ff953c177ff7
facade-service-1 | [facade-service] Enqueuing to counter-queue: msg7
facade-service-1 | INFO: 192.168.65.1:36357 - "POST / HTTP/1.1" 200 OK
facade-service-1 | [facade-service] POST -> logging-service at http://logging-service-3:8083, uuid=f7ffe119-e903-48a2-bbc6-f3c035c840d2
facade-service-1 | [facade-service] Enqueuing to counter-queue: msg8
facade-service-1 | INFO: 192.168.65.1:51788 - "POST / HTTP/1.1" 200 OK
facade-service-1 | [facade-service] POST -> logging-service at http://logging-service-2:8082, uuid=c6280c7d-f660-4d83-a201-54d8e6d9c0f6
facade-service-1 | [facade-service] Enqueuing to counter-queue: msg9
facade-service-1 | INFO: 192.168.65.1:16252 - "POST / HTTP/1.1" 200 OK
facade-service-1 | [facade-service] POST -> logging-service at http://logging-service-2:8082, uuid=deb7bc9d-6450-4479-ad9a-e7e502cb5854
facade-service-1 | [facade-service] Enqueuing to counter-queue: msg10
facade-service-1 | INFO: 192.168.65.1:46362 - "POST / HTTP/1.1" 200 OK
facade-service-1 | [facade-service] GET logs <- http://logging-service-2:8082
facade-service-1 | [facade-service] GET counter <- http://counter-service:8084
facade-service-1 | INFO: 192.168.65.1:27588 - "GET / HTTP/1.1" 200 OK
facade-service-1 | [facade-service] POST -> logging-service at http://logging-service-1:8081, uuid=02d01ab6-6aff-49da-a053-06f139133491
facade-service-1 | [facade-service] Enqueuing to counter-queue: msg11
facade-service-1 | INFO: 192.168.65.1:48911 - "POST / HTTP/1.1" 200 OK
facade-service-1 | [facade-service] POST -> logging-service at http://logging-service-3:8083, uuid=703c4436-1882-4d01-a5b9-23e2ae486192
facade-service-1 | [facade-service] Enqueuing to counter-queue: msg12
facade-service-1 | INFO: 192.168.65.1:35766 - "POST / HTTP/1.1" 200 OK
facade-service-1 | [facade-service] POST -> logging-service at http://logging-service-1:8081, uuid=47f1c735-32a9-460c-bac9-fbd3cd679659
facade-service-1 | [facade-service] Enqueuing to counter-queue: msg13
facade-service-1 | INFO: 192.168.65.1:63279 - "POST / HTTP/1.1" 200 OK
facade-service-1 | [facade-service] POST -> logging-service at http://logging-service-3:8083, uuid=2533f6f7-8d99-464c-8f39-8686f1b2bd07
facade-service-1 | [facade-service] Enqueuing to counter-queue: msg14
facade-service-1 | INFO: 192.168.65.1:36387 - "POST / HTTP/1.1" 200 OK
facade-service-1 | [facade-service] POST -> logging-service at http://logging-service-3:8083, uuid=6b2ca1ad-302f-47bf-b0a6-ea6b925829b4
facade-service-1 | [facade-service] Enqueuing to counter-queue: msg15
facade-service-1 | INFO: 192.168.65.1:42328 - "POST / HTTP/1.1" 200 OK
facade-service-1 | [facade-service] GET logs <- http://logging-service-3:8083
```

Видно:

- підключення до Hazelcast
- для кожного POST: вибір logging-service та постановку в чергу (Enqueuing to counter-queue)
- для GET при недоступному counter-service: Counter-service unavailable: timed out